

Nathanial Schultz

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Experience

- **Advanced Radar Research Center**

Undergraduate Research Assistant

Norman, OK

Feb. 2024 – Present

- Currently involved in implementing software and algorithms for use in integrated sensing and communications (ISAC) research. Future plans are to develop the software to the point of using it for experimental testing and future algorithm development. Additionally, I am researching the 5G NR protocol for future projects in determining the impact of the protocol in ISAC systems.
- Currently conducting literature research into the use of ultra wide-band (UWB) radar in biomedical radar systems. Post- literature review plans include testing algorithms on real-world models for use in a general-purpose biomedical imaging radar.
- Conducted research on developing biomedical phantoms for use in a biomedical imaging radar. My research involved learning how to test the dielectric properties of materials and the viability of using fused deposition modeling (FDM) printing in the development of stable phantom materials.

- **University of Oklahoma IT**

IT Support Specialist & Mentor

Norman, OK

Sept. 2023 – May 2024

- Assisted customers with various IT requests over the phone, in-person, and through online ticketing software.
 - Resolved over 100 tickets of varying difficulty, including software and hardware requests.
 - As a mentor, ensured newer co-workers were assisted with more difficult requests, involved in training new employees, and ensured everyone remained on-track with their work.
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Education

- **University of Oklahoma**

B.S. in Electrical Engineering

Norman, OK

Expected Graduation: May 2026

- *Relevant Coursework:* Digital Signal Processing, Random Signals, Radar Imaging

- **Rose State College**

A.A.S in Cyber Security

Midwest City, OK

July 2022

- Received CompTIA Network+ Certification
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Campus Involvement

- **IEEE Student Branch**

Secretary

Norman, OK

May 2025 – Present

- Leading the process of obtaining a website for our branch and acquiring graduation regalia.
 - Preside over meetings and help ensure officers are accomplishing their duties.
 - Keep track of event attendance and metrics, and ensure events are properly reported to IEEE and the university.
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Core Technical Skills

Languages: MATLAB, $\text{\LaTeX}$, C, Python, Verilog

Software: Intel Quartus/QuestaSim, GNURadio, SolidWorks, Autodesk Fusion